

System Architecture and System Design

Architectural Styles

HomeBased utilizes a layered client-server architectural approach. The front-end (client-side) is built using Bootstrap, HTML, minimal CSS and JavaScript to provide responsive, efficient and dynamic interactions. Client-side requests are processed via routes implemented using Node JS and Express to connect to the appropriate controller and service needed to process the request. Bootstrap was chosen to ensure a wide range of device support.

The back end (server-side) is implemented within a MariaDB server, and MySQL within a normalized relational database. The Database utilizes the built-in B-Tree functionality to optimize server-size processing of client-side requests.

Identifying Subsystems

1) Application

- a. User Interface
- b. A router manipulates the DOM to load the various pages and forms
- c. Pages and forms included:
 - i. Home Page
 - ii. Add Item Form
 - iii. Listed Items
 - iv. Unlisted Items queue
 - v. New Listing Form
 - vi. Sell Item Form

2) Routing System

- a. Receives user requests, then routes them to the appropriate controller
- b. Routes included:
 - i. Items
 - ii. Purchases
 - iii. Stores
 - iv. Platforms
 - v. Listings
 - vi. Transactions

3) Controllers & Services

- a. *Controllers execute API requests to GET data from the database*
 - i. Included Controllers:
 - 1. Items

2. Purchases
3. Stores
4. Platforms

b. *Services execute transactions and POST to the database*

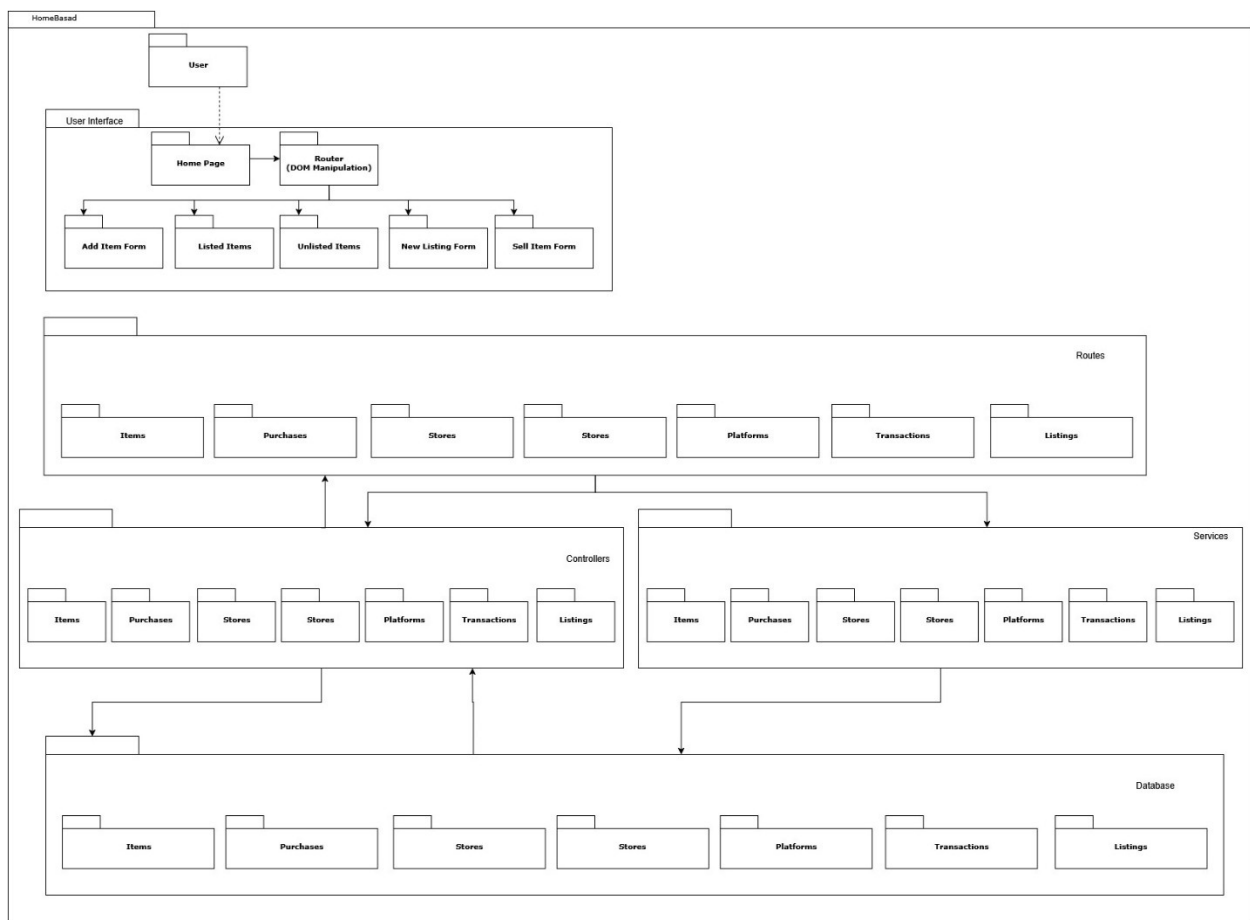
i. Included Services:

1. Item
2. Purchases
3. Stores
4. Platforms
5. Listings
6. Transactions

4) Database

- a. Persistent data storage
- b. Safeguards data integrity via referential integrity constraints

UML Package Diagram



Persistent Data Storage

HomeBased requires data storage that persists beyond a single execution in the system. Data is stored in a relational database and is then accessed or altered using MySQL queries.

The persistent entities stored are as follows:

- 1) Stores
- 2) Purchases
- 3) Items
- 4) Item Specializations
- 5) Platforms
- 6) Listings
- 7) Transactions

Global Control Flow

Execution Orders

Event driven execution is implemented using HTML forms with CSS formatting with event listeners attached via Javascript. Events are then executed via API's modules that utilize the JSON data exchange.

Network Protocol

Communication flows using HTTP protocol that implements Express API to execute database queries. This network protocol was chosen as it aligns with guidance provided by MariaDB for NodeJS implementations utilizing the MariaDB, open-source server and database solution.

Hardware Requirements

HomeBased requires a screen display with a minimal resolution of 1024 x 768 pixels, a broadband internet connection, and a modern web browser that provides modern JavaScript support.

Minimum client hardware requirements:

- 1) Node.js runtime environment
- 2) 1 GB RAM
- 3) 1 CPU Core
- 4) 1GB disk space